Applied Methods - Main Results

Fourier Transforms

Definition (Fourier Transform)

The Fourier transform of $f(x)$ is

$$\tilde{f}(k) = \int_{-\infty}^{\infty} f(x)e^{-ikx}dx$$

Properties of Fourier Transforms

- **Uniqueness:** $f(x) = g(k) \iff \tilde{f}(k) = \tilde{g}(k)$
- **Linearity:** $h(x) = \lambda f(x) + \mu g(x) \implies \tilde{h}(k) = \lambda\tilde{f}(k) + \mu\tilde{g}(k)$
- **Scaling:** $g(x) = f(ax) \implies \tilde{g}(k) = \frac{1}{|a|}\tilde{f}(k/a)$
- **Translation:** $g(x) = f(x-a) \implies \tilde{g}(k) = e^{-ika}\tilde{f}(k)$
- **Frequency Shift:** $g(x) = e^{ik_0x} \implies \tilde{g}(k) = \tilde{f}(k-k_0)$
- **Derivatives:** $g(x) = f'(x) \implies \tilde{g}(k) = ik\tilde{f}(k)$
- **Factors of x:** $g(x) = xf(x) \implies \tilde{g}(k) = i\tilde{f}'(k)$

Definition (Inverse Fourier Transform)

The inverse Fourier transform of $\tilde{f}(k)$ is

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \tilde{f}(k)e^{ikx}dk$$

Theorem (The Fourier Inversion Theorem)

Let $\tilde{f}(k)$ be the Fourier transform of $f(x)$. Then

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \tilde{f}(k)e^{ikx}dk = \begin{cases} f(x) & \text{if } f \text{ is continuous at } x \\ \dfrac{f(x_-) + f(x_+)}{2} & \text{otherwise.} \end{cases}$$

where $f(x_-)$ and $f(x_+)$ are the left and right limits of f at x respectively.

Definition (Convolution)

The convolution of $f(x)$ and $g(x)$ is

$$(f * g)(x) = \int_{-\infty}^{\infty} f(\xi)g(x-\xi)d\xi$$

Note on Convolutions

$$(f * g)(x) = (g * f)(x)$$

Theorem (The Convolution Theorem)

$$h(x) = (f * g)(x) \implies \tilde{h}(k) = \tilde{f}(k)\tilde{g}(k)$$

Dynamical Systems

1D Flows

Definition (Stable and Unstable Eq. Points)

An equilibrium point x^* of $\dot{x} = f(x)$ with $x(0) = x_0$ is stable if $\forall \varepsilon > 0\ \exists \delta > 0$ such that

$$|x - x_0| < \delta \implies |x(t) - x^*| < \varepsilon$$

for all $t > 0$. [1]

An equilibrium point is unstable if it is not stable, obviously.

Definition (Asymptotically Stable Eq. Point)

An equilibrium point is asymptotically stable if it is stable and $\exists\, r > 0$ such that

$$|x(t) - x^*| \to 0 \quad \text{as} \quad t \to \infty$$

for all x_0 such that $|x_0 - x^*| < r$.

Definition (Globally Asymptotically Stable Eq. Point)

An equilibrium point is globally asymptotically stable if

$$|x(t) - x^*| \to 0 \quad \text{as} \quad t \to \infty$$

for all x_0.

Definition (Hyperbolic Eq. Point)

An equilibrium point x^* is hyperbolic if

$$f'(x^*) \neq 0.$$

Else, x^* is a degenerate equilibrium point.

Types of Bifurcation in 1D Flows

- **Saddle node bifurcation** is the mechanism by which equilibrium points are created or destroyed.
- **Transcritical bifurcation** is the mechanism by which equilibrium points change stability.
- **Pitchfork bifurcation** is the mechanism by which equilibrium points are created or destroyed in pairs. A pitchfork bifurcation is **supercritical** if equilibrium points are created when the parameter is greater than the critical value. A pitchfork bifurcation is **subcritical** if equilibrium points are created when the parameter is less than the critical value.
- There is also **hysteresis, fold, and cusp** but who even cares.

[1] Seems like this definition (along with the three below) is not really needed for the exam.

2D Flows

Definition (Equilibrium Point)

A 2D flow $\begin{cases} \dot{x} = f(x,y) \\ \dot{y} = g(x,y) \end{cases}$ has an equilibrium point (x^*, y^*) when $f(x^*, y^*) = g(x^*, y^*) = 0$.

A linear system (or a non-linear system that you can linearise[2]) $\dot{\mathrm{x}} = \mathrm{Mx}$ has an equilibrium point x^* if $\mathrm{Mx}^* = \mathbf{0}$. Recall that $\mathrm{x} = \begin{pmatrix} x & y \end{pmatrix}^T$ and $\mathrm{M} = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix}$ is the Jacobian (evaluated at (x^*, y^*)).

General Solution to Linear 2D Flow

The general solution to $\dot{\mathrm{x}} = \mathrm{Mx}$ where $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with eigenvalues λ_1, λ_2 and corresponding eigenvectors $\mathrm{v}_1, \mathrm{v}_2$ is

$$\mathrm{x}(t) = c_1 e^{\lambda_1 t} \mathrm{v}_1 + c_2 e^{\lambda_2 t} \mathrm{v}_2$$

Method for Second Order ODEs

You can write a second order ODE of the form $\ddot{x} = f(x, \dot{x})$ as a 2D flow by letting $y = \dot{x}$. This leaves us with $\begin{cases} \dot{x} = y \\ \dot{y} = f(x,y). \end{cases}$

Classification of Equilibrium Points for Linear 2D Flows

Let λ_1, λ_2 be the eigenvalues of M with eigenvectors $\mathrm{v}_1, \mathrm{v}_2$ respectively. Let $A = \mathrm{Tr(A)}$ and $D = \det(\mathrm{M})$. It may be helpful to note that $A = \lambda_1 + \lambda_2$ and $D = \lambda_1 \lambda_2$.

- $\lambda_{1,2} \in \mathbb{R} \setminus \{0\}$ where $\lambda_1 \neq \lambda_2$ ($D < A^2/4$ and $D \neq 0$)
  - $\lambda_{1,2} < 0$ ($D > 0$, $A < 0$) corresponds to a **stable node**. Trajectories approach the equilibrium point tangent to the direction of the eigenvector with the smallest (in magnitude) eigenvalue.
  - $\lambda_{1,2} > 0$ ($D > 0$, $A > 0$) corresponds to an **unstable node**. All trajectories move away from the origin. I think that trajectories approach the direction of the eigenvector with the smallest eigenvalue again, but idk.
  - $\lambda_1 < 0 < \lambda_2$ ($D < 0$) corresponds to a **saddle node**. Only trajectories beginning on the line generated by v_1 (stable manifold) will approach the equilibrium point. Other trajectories approach the direction of v_2 (unstable manifold).
- $\lambda_{1,2} \in \mathbb{C} \setminus \mathbb{R}$ where $\lambda_1 \neq \lambda_2$ ($D > A^2/4$ and $D \neq 0$)
  - Re($\lambda_{1,2}$) < 0 ($A < 0$) corresponds to a **stable spiral**. Trajectories in to the origin.
  - Re($\lambda_{1,2}$) > 0 ($A > 0$) corresponds to an **unstable spiral**. Trajectories spiral out from the origin.
  - Re($\lambda_{1,2}$) $= 0$ ($A = 0$) corresponds to a **center**. Trajectories are stable and periodic "around" the origin. Note they are not attracted to or repelled by the origin.

[2] For boundary cases this doesn't always work :(. Minimal guidance has been given on how to handle these, so if it comes up in the exam we've just gotta get good and figure it out I guess.

- $\lambda_1 = \lambda_2 = \lambda \neq 0$ ($D = A^2/4$ and $D \neq 0$)
  - Two linearly independent eigenvectors corresponds to a **stable** ($\lambda < 0$) or **unstable** ($\lambda > 0$) **star**. Trajectories approach the origin in straight lines.
  - Only one linearly independent eigenvector corresponds to a **stable** ($\lambda < 0$) or **unstable** ($\lambda > 0$) **degenerate node**. Trajectories approach or are repelled by the origin tangent to the direction of the eigenvector.
- $\lambda_1 = 0$ ($D = 0$) corresponds to a **line of equilibrium points**.
  - $\lambda_2 \neq 0$ means the lines are either **stable** ($\lambda_2 < 0$) or **unstable** ($\lambda_2 > 0$). Trajectories are attracted to or repelled by this line parallel to the direction of v_2.
  - $\lambda_2 = 0$ means that the equilibrium points do not attract or repel trajectories. If M is the zero matrix, the whole plane is full of equilibrium points.

Figure 1 shows what the phase portraits should look like and their location in the A-D plane.[3] Notice:

- The green area above the $D = A^2/4$ curve corresponds to strictly complex eigenvalues (hence the rotation-like properties of the phase portraits), and the pink area below the curve corresponds to real eigenvalues.
- The left half plane contains only stable nodes, and the right half plane contains only unstable nodes, ignoring saddle nodes.
- The stars and degenerate nodes are positioned somewhat arbitrarily on the $D = A^2/4$ curve, as you need to check the eigenvectors in order to determine which case you are in.

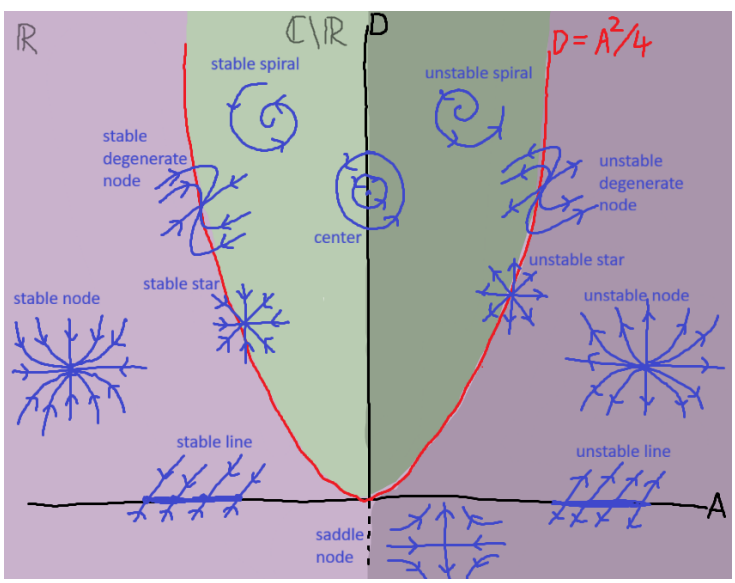

Figure 1: Classification of equilibrium points for a 2D linear flows.

[3] I drew Figure 1 in MS Paint because I didn't like any pre-existing versions, so it looks a bit shit. Although, it did lead me to discover that you can memorise a great deal about the classification quite easily by just drawing Figure 1 a few times.

Limit Cycles

Definition (Limit Cycle)

A limit cycle is an isolated closed trajectory. That is, no neighbouring trajectories are also closed.

Definition (Stable and Unstable Limit Cycles)

If all neighbouring trajectories approach the limit cycle, it is stable. Else, it is unstable (or half-stable, whatever that means).

Note on Limit Cycles

Limit cycles cannot occur in linear systems.

Definition (Gradient System, Potential Function)

A system $\begin{cases} \dot{x} = f(x, y) \\ \dot{y} = g(x, y) \end{cases}$ is a gradient system if there exists a continuously differentiable scalar function[4] F such that $\dot{\mathrm{x}} = -\nabla F(\mathrm{x})$ where $\nabla F(\mathrm{x}) = \begin{pmatrix} F_x & F_y \end{pmatrix}^T$. We call F a potential function.

Definition (Lyapunov Function)

Consider a 2D flow with equilibrium point (x^*, y^*). A function $V(\mathrm{x})$ is a Lyapunov function if and only if the following hold:

- V is continuously differentiable.
- $V(x^*, y^*) > 0$ for all $(x, y) \neq (x^*, y^*)$. That is, V is strictly positive,
- $\dot{V}(x^*, y^*) < 0$ for all $(x, y) \neq (x^*, y^*)$. That is, V is strictly decreasing.

Result on Lyapunov Functions

If a system with equilibrium point (x^*, y^*) has a Lyapunov funcion V, then (x^*, y^*) is globally asymptotically stable.

Result(s) on Gradient Systems, Lyapunov Functions, and Limit Cycles

If a system is gradient system, or has a Lyapunov function, then the system has no closed orbits. In particular, the system has no limit cycles.

[4] Continuously differentiable just means that F has continuous derivatives. A scalar function just returns a real number e.g. the energy of a system.

Poincaré-Bendixson Theorem

Suppose that

- R is a closed, bounded subset of the plane.
- $\dot{x} = f(x, y)$ and $\dot{y} = g(x, y)$ are continuously differentiable on an open subset containing R.
- R does not contain any equilibrium points.
- There is a trajectory C that stays within R for all t.

Then either C is a closed orbit, or C spirals towards a closed orbit as $t \to \infty$. In any case, there is a closed orbit[5] in R.

Bifurcation in 2D

All definitions for 2D bifurcation points are basically the same in 1D.

Definition (Hopf Bifurcation)

- **Supercritical** Hopf bifurcation occurs when a stable node becomes unstable, and a limit cycle is created around the unstable node.
- **Subcritical** Hopf bifurcation occurs when an unstable node becomes stable, and a limit cycle around the unstable node is destroyed (collapses into the stable node).

This is roughly equivalent to two complex conjugate eigenvalues of the Jacobian crossing the imaginary axis. That is, Hopf bifurcation occurs at the point where the eigenvalues are purely imaginary[6].

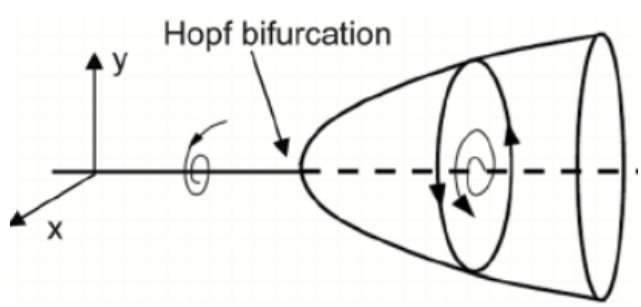

Figure 2: Hopf Bifurcation Diagram. Read from left to right for supercritical, right to left for subcritical.

The End.

[5] I don't think this determines whether or not there is a limit cycle... Reasoning: the trajectories around centers (for linear systems) are not limit cycles, but satisfy the above conditions when we take R to be any closed disc.

[6] There are a load of other conditions too but they aren't covered in the module and are probably just there to exclude weird pathological cases.